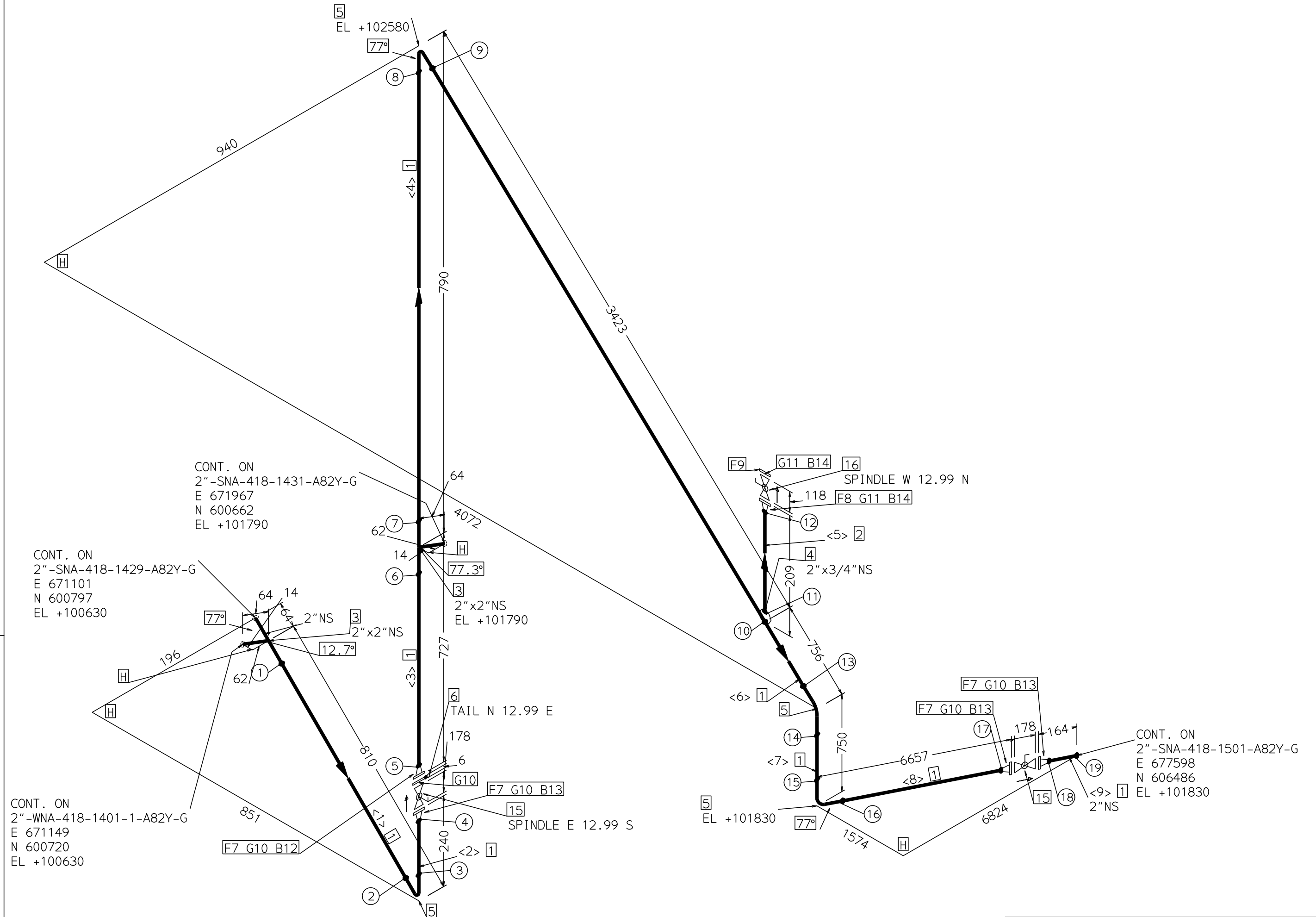


| PT<br>NO                      | COMPONENT DESCRIPTION                                                    | N.S.<br>(INS) | ITEM CODE          | QTY   |
|-------------------------------|--------------------------------------------------------------------------|---------------|--------------------|-------|
| <b>PIPE</b>                   |                                                                          |               |                    |       |
| 1                             | PIPE B-36.19 BE SEAMLESS 40S                                             | 2             | P13D217Z00913ZZZZ  | 13.3M |
| 2                             | PIPE B-36.19 PE SEAMLESS 80S                                             | 3/4           | P13D277Z00514ZZZZ  | 0.1M  |
| <b>FITTINGS</b>               |                                                                          |               |                    |       |
| 3                             | T.EQUAL B-16.9 BW 40S                                                    | 2 x 2         | WEGC84ZZ00913ZZZZ  | 2     |
| 4                             | WELDOLET MSS-SP97 BW 40S 80S                                             | 2 x 3/4       | YWSAF4ZZ009130514  | 1     |
| 5                             | ELBOW.90 B-16.9 BW 1.5D 40S                                              | 2             | WAGC84Z100913ZZZZ  | 4     |
| 6                             | FLNG.FIG.8 ASME-B16.48 150 FF/125AARH                                    | 2             | FGKG521Z009ZZZZZZ  | 1     |
| <b>FLANGES</b>                |                                                                          |               |                    |       |
| 7                             | FLNG.WN B-16.5 150 RF/125AARH 40S                                        | 2             | FWCG527Z00913ZZZZ  | 4     |
| 8                             | FLNG.WN B-16.5 150 RF/125AARH 80S                                        | 3/4           | FWCG527Z00514ZZZZ  | 1     |
| 9                             | FLNG.BLIND B-16.5 150 RF/125AARH                                         | 3/4           | FBGCG527Z005ZZZZZZ | 1     |
| <b>ERECTION MATERIALS</b>     |                                                                          |               |                    |       |
| PT<br>NO                      | COMPONENT DESCRIPTION                                                    | N.S.<br>(INS) | ITEM CODE          | QTY   |
| <b>GASKETS</b>                |                                                                          |               |                    |       |
| 10                            | GASKET B-16.20-ANSI B16.5 SPIRAL 150                                     | 2             | GK6CF72Z009ZZZZZZ  | 5     |
| 11                            | GASKET B-16.20-ANSI B16.5 SPIRAL 150                                     | 3/4           | GK6CF72Z005ZZZZZZ  | 2     |
| <b>BOLTS</b>                  |                                                                          |               |                    |       |
| 12                            | 95 MM LONG BOLT.STUD WITH 2 NUTS B-18.2 A193 GR.B7(HDG)/ A194 GR.2H(HDG) | 5/8           |                    | 4     |
| 13                            | 83 MM LONG BOLT.STUD WITH 2 NUTS B-18.2 A193 GR.B7(HDG)/ A194 GR.2H(HDG) | 5/8           |                    | 12    |
| 14                            | 64 MM LONG BOLT.STUD WITH 2 NUTS B-18.2 A193 GR.B7(HDG)/ A194 GR.2H(HDG) | 1/2           |                    | 8     |
| <b>VALVES / IN-LINE ITEMS</b> |                                                                          |               |                    |       |
| 15                            | VLV.BALL SHEET 543DR                                                     | 2             | 543DRZZZ009ZZZZZZ  | 2     |
| 16                            | VLV.BALL SHEET 543DR                                                     | 3/4           | 543DRZZZ005ZZZZZZ  | 1     |

| PT<br>NO                      | COMPONENT DESCRIPTION                   | N.S.<br>(INS) | ITEM CODE | QTY              |    |
|-------------------------------|-----------------------------------------|---------------|-----------|------------------|----|
| <u>GASKETS</u>                |                                         |               |           |                  |    |
| 10                            | GASKET B-16.20-ANSI B16.5 SPIRAL        | 150           | 2         | GK6CF72Z009ZZZZZ | 5  |
| 11                            | GASKET B-16.20-ANSI B16.5 SPIRAL        | 150           | 3/4       | GK6CF72Z005ZZZZZ | 2  |
| <u>BOLTS</u>                  |                                         |               |           |                  |    |
| 12                            | 95 MM LONG BOLT.STUD WITH 2 NUTS B-18.2 | 5/8           |           |                  | 4  |
|                               | A193 GR.B7(HDG)/ A194 GR.2H(HDG)        |               |           |                  |    |
| 13                            | 83 MM LONG BOLT.STUD WITH 2 NUTS B-18.2 | 5/8           |           |                  | 12 |
|                               | A193 GR.B7(HDG)/ A194 GR.2H(HDG)        |               |           |                  |    |
| 14                            | 64 MM LONG BOLT.STUD WITH 2 NUTS B-18.2 | 1/2           |           |                  | 8  |
|                               | A193 GR.B7(HDG)/ A194 GR.2H(HDG)        |               |           |                  |    |
| <u>VALVES / IN-LINE ITEMS</u> |                                         |               |           |                  |    |
| 15                            | VLV.BALL SHEET 543DR                    | 2             |           | 543DRZZZ009ZZZZZ | 2  |
| 16                            | VLV.BALL SHEET 543DR                    | 3/4           |           | 543DRZZZ005ZZZZZ | 1  |

| PIECE NO | CUT LENGTH | N. S. (INS) | PIECE NO | CUT LENGTH | N. S. (INS) |
|----------|------------|-------------|----------|------------|-------------|
| <1>      | 671        | 2           | <2>      | 100        | 2           |
| <3>      | 600        | 2           | <4>      | 651        | 2           |
| <5>      | 100        | 3/4         | <6>      | 4027       | 2           |
| <7>      | 598        | 2           | <8>      | 6517       | 2           |
| <9>      | 100        | 2           |          |            |             |



| WD | SW | FIT | WLDR | VIS | DP | DP  | RAD. | SR.  | HRD |
|----|----|-----|------|-----|----|-----|------|------|-----|
| NO | FW | UP  | NO   | ACC | RT | FNL | DATE | DATE | NSS |
| 1  | S  |     |      |     |    |     |      |      |     |
| 2  | S  |     |      |     |    |     |      |      |     |
| 3  | S  |     |      |     |    |     |      |      |     |
| 4  | S  |     |      |     |    |     |      |      |     |
| 5  | S  |     |      |     |    |     |      |      |     |
| 6  | S  |     |      |     |    |     |      |      |     |
| 7  | S  |     |      |     |    |     |      |      |     |
| 8  | S  |     |      |     |    |     |      |      |     |
| 9  | S  |     |      |     |    |     |      |      |     |
| 10 | S  |     |      |     |    |     |      |      |     |
| 11 | S  |     |      |     |    |     |      |      |     |
| 12 | S  |     |      |     |    |     |      |      |     |
| 13 | S  |     |      |     |    |     |      |      |     |
| 14 | S  |     |      |     |    |     |      |      |     |
| 15 | S  |     |      |     |    |     |      |      |     |
| 16 | S  |     |      |     |    |     |      |      |     |
| 17 | S  |     |      |     |    |     |      |      |     |
| 18 | S  |     |      |     |    |     |      |      |     |
| 19 | S  |     |      |     |    |     |      |      |     |

|             |                   |
|-------------|-------------------|
| INCHDIA: 40 | INCHMETER: 30.025 |
|-------------|-------------------|

CONSULTANT:  **ENGINEERS INDIA LIMITED**  
**NEW DELHI**

**MUNITIONS INDIA LIMITED**  
**HEF KHADKI-PUNE, INDIA**

FILE: AFCIS075A2

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CUMULATIVE CENTER LINE CURVILINEAR LENGTHS ( INCLD. VALVES)

PIPE NS (INS):3/4 2

CL LENGTH (M):0.3 14.6

|     |      |           |    |     |     |     |       |
|-----|------|-----------|----|-----|-----|-----|-------|
|     |      |           |    |     |     |     |       |
|     |      |           |    |     |     |     |       |
|     |      |           |    |     |     |     |       |
|     |      |           |    |     |     |     |       |
| NO. | DATE | REVISIONS | BY | CHK | MTR | STR | APPVD |

|                                                                     |  |              |
|---------------------------------------------------------------------|--|--------------|
| PROJECT NAME : MIL PROJECT<br>TNT PLANT, HEF KHADKI PUNE,<br>INDIA. |  |              |
| UNIT NAME : SNA, WNA & OLEUM TANKS                                  |  |              |
| JOB NO : B889                                                       |  | ZONE : G     |
| LINE NUMBER                                                         |  | REV          |
| 2"-SNA-418-1430-A82Y-G                                              |  | -1           |
| UNIT : 418A                                                         |  | SHEET: 1 / 1 |